Research, Experience, and Project Portfolio

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My work spans applied mathematics, biomedical AI, public health, scientific software, and collaborative community projects. This portfolio distinguishes published research, research manuscripts and software, project design, and coursework. Public materials are linked where available; clinical records, internal business materials, and third-party source collections are not distributed through this profile.

Research and scientific software

Structured PCV vaccination modeling

Research manuscript and open-source computational framework

Developed a framework for vaccination-program analysis that connects continuous-time Markov dynamics, cohort ordinary differential equations, and age-calendar transport formulations under matched conditions. The work combines mathematical derivations with separately evaluated pneumococcal-vaccine product scenarios.

- Organized **144 scenario configurations**, traceable parameter and source ledgers, retained simulation outputs, and figure-to-script mappings.
- Prepared a research manuscript, six main figures, two main tables, and a versioned reproducibility archive.
- Documented solver checks, accounting consistency, portability diagnostics, and interpretation boundaries.

The PCV application is a methodological and conditional scenario study. Its computational checks are distinct from empirical calibration and clinical or policy validation.

Code and documentation · Aligned v1.0.1 release

Vietnam vaccination-modeling exploration

Earlier computational modeling project

Explored vaccination analysis through Markov and ODE implementations, QALY calculations, and visualization scripts. This earlier repository records the development of my interest in mathematical public-health modeling alongside the more extensively documented structured PCV framework.

Repository

Radiotherapy Physics RAG and evidence workflows

Open-source retrieval software and related research prototypes

Built local evidence-retrieval tools for question answering over radiotherapy physics reports, with report-aware retrieval, structured citations, provenance metadata, and an MCP interface.

- Developed workflows for document parsing, chunking, retrieval, navigation, and evidence inspection.
- Organized reproducible evaluation procedures and source-grounded output contracts.
- Extended the research toward bounded literature-workflow execution, evidence ledgers, failure retention, and recovery.
- Investigated limitations in automated scoring, including cases in which surface evidence matching does not capture factual polarity.

The public QA software and later research prototypes represent different development stages. Automated benchmark results are research measures rather than clinical validation; the controlled research-workflow evaluation did not establish superiority over its literature-workflow baseline.

Public QA repository

Medical imaging and representation learning

PET/CT research engineering and Summer Research Scholar work

Developed medical-imaging preprocessing and feature-extraction workflows covering DICOM conversion, organ segmentation, PET-to-CT spatial resampling, and aggregation of foundation-model embeddings.

- Implemented resumable processing and output-integrity checks.
- Worked with image preprocessing, patch-level feature extraction, and aggregation across anatomical masks and study records.
- Pursued medical-imaging research under Zhenyu Yang through the Duke Kunshan University Summer Research Scholar program.
- Prepared a subsequent research proposal on uncertainty-guided abdominal CT segmentation, prediction-disagreement visualization, and reproducible experiment design.

The uncertainty-guided segmentation extension is presented as a research proposal. Patient-level images and feature tables remain outside this public portfolio.

Radiomics and dosiomics for radiation-pneumonitis research

Research manuscript

A Dual Radiomic and Dosiomic Filtering Technique for Locoregional Radiation Pneumonitis Prediction in Breast Cancer Patients examines an imaging-and-dose feature-analysis approach to radiation-pneumonitis prediction. This manuscript is a distinct research topic from my radiotherapy-physics RAG software and is presented here as manuscript work rather than a published article.

Protein-corona and molecular data engineering

Data integration, molecular processing, and research-quality control

Developed traceable workflows for protein-corona research by harmonizing five source collections while retaining assay, species, sample-role, and measurement distinctions.

- Organized **1,346 source-sample records** and source-specific protein or protein-group information.
- Processed a molecular-structure collection containing **50,547 records**, including **10,962 phospholipid records**.
- Implemented structural fragmentation and reconstruction checks, source identity tracking, and explicit handling of unresolved mappings.
- Preserved distinctions between identification-only records and quantitative abundance measurements.

This work focuses on data engineering and structural quality control. Harmonized identifiers do not imply that all measurement scales are directly comparable.

AI-assisted cancer-screening health economics

Research design and methodological development

Worked on the design of a health-economic evaluation framework for AI-assisted hepatobiliary and pancreatic cancer screening. The project considers screening pathways, risk stratification, discrete-time state-transition modeling, costs and QALYs, and uncertainty-analysis plans.

The current portfolio contribution is methodological design and research documentation; planned simulation sizes and sensitivity analyses are not presented as completed experimental results.

AI-assisted circuit and PCB generation

Engineering prototype

Prototyped circuit-generation workflows using Python, SKiDL, and KiCad, with buck-converter examples, schematic and netlist generation, and toolchain-compatibility documentation. The work includes generated design artifacts and demonstrations of the software workflow, without a claim of independently verified hardware performance.

Multi-vaccine cost-effectiveness and financing

Signature Work research proposal

Proposed a unified framework for comparative vaccine cost-effectiveness and financing analysis in low- and middle-income countries. The planned extension connects prior vaccine modeling with multi-vaccine comparison, uncertainty, and alternative financing arrangements.

Undergraduate innovation research

Soul Corridor: death education and end-of-life care

Undergraduate innovation research project · March 2024 · Mentor: Meifang Chen

Pursued **"Soul Corridor: Illuminating End-of-Life Care in China through Death Education,"** a project on death education and attitudes toward palliative and end-of-life care. The research combines literature synthesis, questionnaire-based investigation, and the communication of sensitive public-health topics. My work focused on organizing the research question, coordinating project tasks, and connecting evidence review with survey design and analysis.

Public-data liver-fibrosis risk benchmark

Undergraduate innovation research project · March 2025 · Mentor: Liqi Ren

Explored liver-fibrosis risk modeling using NHANES data, with attention to temporal validation and comparisons between established risk scores and statistical or machine-learning approaches. The project considers FIB-4 and APRI alongside logistic regression, penalized regression, and random forests, with an emphasis on transparent reporting, model documentation, and reproducible evaluation.

These undergraduate projects are presented as research experience rather than published or clinically validated diagnostic systems.

Conference publication

Mengkai Dai, Cheng Peng, Rui Li, and **Ruimeng Zhou**. **Online Electrochemical Impedance Spectroscopy Identification in Reconfigurable Battery Systems Using Switching Harmonic Excitation.** 2025 IEEE 3rd International Power Electronics and Application Symposium (PEAS). DOI: 10.1109/PEAS66638.2025.11403555

Industry experience

New Oaks AI - Market Analyst Intern (May-August 2025). AI/SaaS competitor research, social-media analysis, creator-outreach research, and content planning. Prepared recurring analytical materials to support market and communication decisions.

Jade Capital - Market Analyst Intern (November 2024-May 2025). Industry and value-chain research, PE/VC case analysis, and scenario-based financial analysis. Prepared internal research reports and supporting analytical materials.

China Eastern Airlines, Shandong Branch - Route Data Analysis Intern (June 12-August 11, 2024). Completed an internship at the Public Finance Center, contributing to route data analysis through on-site and online work.

These summaries describe research and analytical responsibilities. Internal business documents and unverified growth or return claims are not reproduced here.

Teaching, mentoring, and university research support

Peer Mentor & Peer Tutor, Academic Advising (September 2024-May 2025). Academic transition support, mentoring, and course-learning assistance.

Research/Teaching Assistant, Edge Intelligence Lab (November-December 2024). Research and teaching support within Natural and Applied Sciences.

Research Assistant, Environment Research Center (July-August 2024). Research data collection, preparation, and standardization for empirical analysis.

Chinese Student Services Assistant (September 2023-May 2024). Student-facing administrative and campus-service support.

CSS Military Training Assistant (May-June 2024). Coordination and student support for military-training activities.

Library Student Worker (February-April 2024). Library operations and student-service support.

Additional teaching work includes introductory computer-science learning support and mathematical-modeling project guidance. I completed **Tutor Essentials** training in August 2025.

Student leadership, service, and communication

Student Leaders Board - IDI Committee

Contributed as a Board Member to intercultural engagement, diversity, and inclusion initiatives. Worked with other student leaders on Midnight Breakfast events, a Language Partner Program, and the Interwoven campaign, combining event planning, communication, and delivery.

The 2025 annual report records more than 300 attendees at each of two Midnight Breakfast events, more than 100 Language Partner Program participants, and more than 300 Interwoven participants. These are team-level event figures, not counts of unique individuals reached across all activities.

AALAU 2025 - interdisciplinary dialogue

Invited as a student panelist for **“Voice Matters: Integrating Humanities and Sciences in Transdisciplinary Learning”** at the Alliance of Asian Liberal Arts Universities Annual Presidents’ Forum hosted by Duke Kunshan University, as listed in the official program.

Official conference program

Student-organization administration and academic programming

- **Club Council:** developed reusable communication templates and onboarding guidance covering club applications, supporting materials, review stages, and student-organization administration.
- **Zu Chongzhi Center student committee:** co-developed the AY 2026-27 budget and programming proposal for mathematical-modeling activities, Pi Day, and research-oriented events.
- **Campus engagement:** contributed to student-facing orientation, admissions, and campus-event activities, with an emphasis on communication and coordination.

U-Corp green education and collaborative projects

Worked with a DKU student team on green education at county-level middle schools, connecting photovoltaics and water-purification concepts with interdisciplinary teaching. Project activities included literature review, preparatory interviews, lesson development, field-based teaching, and reflection. Related U-Corp work expanded my experience in collaborative, externally oriented student projects.

Music, media, and finance communication

- **Pop Music Club:** served on the Executive Board and supported student music programming and collaboration among campus performers.
- **Band and orchestra activities:** participated in music-related collaborative work and performance preparation.
- **Finance Club:** authored a published recap of a campus talk on financial inequality and contributed to finance-related student programming. Published article
- **Bilingual communication:** developed event-hosting scripts and presentation materials for student and community activities.

Modeling competitions and innovation projects

MCM 2026 - smartphone battery modeling

Contributed to a team project, **"A Coupled Electro-Thermal-Aging Model for Smartphone Battery Life Prediction and Optimization."** The work integrates an equivalent-circuit representation, power-demand components, a two-node thermal model, aging mechanisms, and numerical experiments implemented in notebooks.

L'Oréal Brandstorm

Participated in **L'Oréal Brandstorm 2024**, developing experience in collaborative innovation and concept communication. Subsequent Brandstorm-related project materials extend this interest in turning research and creative ideas into concise presentations. Participation is distinguished from an award or ranking.

AI-generated-content exploration

Developed presentation, script, and concept materials for AI-generated-content activities, extending my interest in the communication and practical use of generative tools.

Selected coursework and technical projects

Implementation and Experiments for Preconditioned GMRES (Numerical Analysis, 2025). Python implementation using CSR sparse storage, Arnoldi iteration, modified Gram-Schmidt, and Givens rotations; Jacobi and Gauss-Seidel preconditioning; convergence and orthogonality diagnostics; a report and notebook.

From Studio to Live (Signal Processing and Applied Mathematics, Duke University, Spring 2026). Team project with Minxue Lin and Simon Cao; equalization, modulation, reverberation, and related audio effects; waveform, FFT, and spectrogram comparisons.

Donor Hyperglycemia and Liver Graft Dysfunction (Statistical-analysis coursework, Duke University). R/Quarto exploratory analysis, adjusted logistic regression, odds-ratio reporting, and predicted-probability visualization for an observational health-data question.

Java data-structures database prototype (Computer Science coursework). Collaborative report with Xiaoyi Zhu; binary search trees, heaps, hash maps with quadratic probing, record retrieval, sorting, and word-frequency queries.

COVID-19 compartmental modeling (Mathematical Modeling coursework). Exploratory SIR modeling and intervention-scenario analysis, providing experience in translating assumptions into dynamical systems.

Bridging the Immunization Divide (Global Health coursework, Duke University). Policy brief and communication plan on vaccine-access inequalities across Asian income settings, with audience-specific research communication.

INFINITY MUSIC (Moving-image practice coursework). Documentary production involving a student band, interviews, rehearsal footage, sound recording, and production reflection.

Film remakes and national identity (WOC219 coursework). Comparative critical writing on Infernal Affairs and The Departed, examining cultural translation and institutional recognition.

Heritage-resource conservation in Jiangsu (Collaborative social-research coursework). Team report on conservation, public engagement, research framing, and data-informed recommendations.

My broader academic preparation includes calculus and linear algebra; probability and statistics; differential equations and dynamical systems; real and complex analysis; topology; computer science and data structures; physics and chemistry; economics and global health; ethics; cultural analysis; and academic communication. Course materials and reference collections are kept distinct from completed original projects.

Academic recognition and training

- **Undergraduate Entrance Scholarship**, Duke Kunshan University.
- **Summer Research Scholar selection**, 2025, with a further summer-research appointment in 2026.
- **Dean's List with Distinction**, Spring 2024 and Fall 2025; **Dean's List**, Spring 2026.
- **Outstanding Peer Tutor nomination** and **Outstanding Peer Mentor nomination**, academic year 2024-25.
- **Tutor Essentials**, certificate of completion, August 2025.

Nominations are identified separately from awards. Research participation and appointment are distinct from publication.

Extended technical toolkit

Python; R; scientific computing; mathematical modeling; numerical analysis; statistical modeling; data analysis; medical image processing; reproducible research; machine learning; health economics; ordinary and partial differential equations; Monte Carlo simulation; signal processing; Git; NumPy; pandas; SciPy; scikit-learn; PyTorch; Quarto; LaTeX; DICOM; NIfTI; retrieval-augmented generation.

Research practice

I aim to make research reusable through clear problem definitions, source and parameter provenance, versioned code, retained failures, documented computational checks, and readable technical communication. I distinguish a working implementation from a validated scientific claim, and a research manuscript from a peer-reviewed publication.

Research interests: biomedical AI; medical image processing; evidence-grounded language-model workflows; mathematical public-health modeling; health economics; numerical methods; reproducible scientific computing.

Tools: Python, NumPy, SciPy, pandas, Matplotlib, R, tidyverse, Quarto, Git, and domain-specific imaging and circuit-design tools.